

RESEARCH PROFILE

PhD researcher working primarily on open quantum systems and non-equilibrium quantum dynamics. Research interests include quantum many-body dynamics, relaxation and transport, dissipation and decoherence, non-Hermitian physics, statistical mechanics, quantum information, quantum simulation, and connections with machine learning.

EDUCATION AND AFFILIATION

Aug. 2022 - Present	PhD Candidate in Physics Institute of Physics, Chinese Academy of Sciences · Beijing, China
Aug. 2018 - Jun. 2022	B.Sc. in Physics, Honors Program in Science Jilin University · Jilin, China GPA: 3.88/4.00 (93.2/100).

RESEARCH INTERESTS

- Non-equilibrium quantum dynamics
- Quantum many-body physics
- Statistical mechanics
- Open quantum systems
- Non-Hermitian physics
- Quantum information and simulation

ONGOING RESEARCH PROJECTS

OYA-OQS: Liouvillian Lieb-Schultz-Mattis Constraints in Open Quantum Systems ↗

Investigating spectral constraints in translationally invariant open quantum many-body systems with strong $U(1)$ symmetry. The project combines analytical Weyl-sequence arguments with momentum-resolved exact diagonalization and tensor-network simulations of dissipative spin chains.

ssb-qme: Symmetry-Enforced Strong Quantum Mpemba Effect ↗

Developing a symmetry-based mechanism for strong quantum Mpemba effects in symmetry-broken open systems. Current work studies discrete parity and continuous $U(1)$ symmetries through Liouvillian analysis and quantum-master-equation simulations.

PUBLICATIONS AND PREPRINTS

- 2026 **Geometry-adaptive formulation of non-Bloch bands in arbitrary dimensions and spectral instability** ↗
Ze-Yu Xing, Yuncheng Xiong, and Haiping Hu
Communications Physics 9, 127 (2026)
Ze-Yu Xing and Yuncheng Xiong contributed equally.
- 2025 **Universal spreading dynamics in quasiperiodic non-Hermitian systems** ↗
Ze-Yu Xing, Shu Chen, and Haiping Hu
Physical Review B 111, L180203 (2025)
- 2024 **Non-Hermitian skin effect in arbitrary dimensions: non-Bloch band theory and classification** ↗
Yuncheng Xiong, Ze-Yu Xing, and Haiping Hu
Preprint, arXiv:2407.01296 (2024)

HONORS AND AWARDS

- 2020 **Ny Tsi-Ze Scholarship**
Institute of Physics, Chinese Academy of Sciences
- 2020 **Honorable Mention, Mathematical Contest in Modeling**
Consortium for Mathematics and Its Applications
- 2019 **National Scholarship**
Ministry of Education, People's Republic of China
- 2019 **Outstanding Student of Jilin University**
Jilin University